Building a 500K+ lead scraping pipeline for Mexico in two weeks

The most cost-effective path to scraping all Mexican businesses combines Outscraper for Google Maps ($3,200 for 3M records), INEGI's free DENUE API (5M+ businesses), and Apify for social media — all flowing into Supabase at $75/month. Total one-time cost for a comprehensive 3M+ lead database runs approximately **$14,000–$19,000**, with monthly maintenance under $500. The two-week timeline is aggressive but achievable if you prioritize the highest-value data sources and defer enrichment to a rolling process. Mexico's legal framework permits scraping publicly available business data from incorporated entities, but outbound WhatsApp cold messaging carries very high ban risk.

Google Maps dominates Mexican business data — and Outscraper wins on price

Mexico has an estimated **4.9 million registered business establishments** (per INEGI census data), of which roughly **3–5 million** appear on Google Maps. For scraping this entire universe, three tools compete — but one stands far above.

Outscraper offers the best price-to-value ratio at massive scale. Their tiered pricing drops to **$1 per 1,000 records** above 100,000 records, outscraper IGLeads making a full 3M-record scrape cost approximately **$3,200** as a one-time payment. The platform extracts **40+ data fields** including phone number, address, business category, rating, reviews count, hours, website, and Google Place ID. Outscraper Their grid-scanning system uses Mexico's 5-digit postal code system to bypass Google Maps' 120-result-per-query limit, with built-in deduplication via Place ID. Outscraper Export formats include CSV, Excel, JSON, and Parquet. Outscraper Spanish language is natively supported.

Apify's Google Maps Scraper charges $0.004 per place (base) apify Apify and climbs to $0.010 per place with filters, additional details, and contact enrichment. For 3M records, this means **$12,000–$30,000** depending on enrichment level — 4x to 9x more expensive than Outscraper. Apify's advantage is its ecosystem integration with Instagram and Facebook scrapers on the same platform, Hack'celeration plus native webhooks for automation.

SerpAPI charges a recurring **$3,750+/month** at enterprise scale serpapi SerpApi and requires building your own query loop and pagination logic. Each API call returns only one page (~20 results), demanding 300,000+ calls for 3M businesses. The recurring cost model makes it the least attractive option for a one-time bulk scrape.

Tool	Cost for 3M records	Payment model	Ease of use
Outscraper	**$3,200**	One-time, pay-as-you-go	★★★★★ No-code UI
Apify	$12,000–$30,000	One-time per-event	★★★★☆ JSON config
SerpAPI	$4,300+/month recurring	Monthly subscription	★★★☆☆ Requires coding

To ensure comprehensive coverage, structure queries as a **category × city matrix**: 25 business categories in Spanish (restaurante, dentista, abogado, mecánico, veterinaria, gimnasio, etc.) across 500+ Mexican municipalities, subdividing dense metros like CDMX, Guadalajara, and Monterrey by postal code. This generates roughly **15,000–20,000 queries** that will produce 8–10M raw results, deduplicating to 3–5M unique businesses via Google Place ID. Outscraper

---

DENUE is Mexico's hidden goldmine — and it's completely free

The single most valuable discovery in this research: **INEGI's DENUE (Directorio Estadístico Nacional de Unidades Económicas)** provides a free API with data on **5+ million active business establishments** nationwide. INEGI This government-maintained directory includes business name, legal name (razón social), economic activity classification, employee count range, full address with postal code, phone number, email, website, and GPS coordinates. INEGI

Free Python (`INEGIpy`), GitHub R (`inegiR`), CRAN and Node.js libraries exist for API access. GitHub The only limitation is that phone/email/website fields are not always populated — many micro-businesses don't report these details. But as a foundation layer for business names, addresses, categories, and locations, DENUE is unmatched. **Start here, then enrich with Google Maps scraping.**

Other Mexican directories worth scraping:

- **Doctoralia Mexico** (doctoralia.com.mx): **180,000+ verified healthcare practitioners** including doctors, dentists, psychologists, and nutritionists. App Store Apify has a dedicated "Doctoralia Data Extractor" actor. Apify Data includes specialty, clinic addresses, phone numbers, patient reviews, insurance accepted, and consultation fees. Actowiz Solutions WhatsApp and email are not directly exposed.
- **Sección Amarilla** (seccionamarilla.com.mx): Still active with **170,000+ advertisers** Google Cloud and 9–10 million monthly visits. DeDinero Now owned by Telmex and operating as a digital marketing agency. DeDinero Useful but redundant if Google Maps is already scraped.

- **TripAdvisor Mexico: 103,000+ restaurants** listed, strong in tourist zones. Valuable specifically for hospitality businesses.
- **TopDoctors Mexico** (topdoctors.mx): **~12,900 curated specialists** Health Connections — smaller but higher-quality than Doctoralia.
- **Yelp Mexico**: Poor coverage. Reviews skew toward English-speaking tourists. Not worth the effort.

Social media scraping unlocks WhatsApp and Instagram contacts

Facebook is Mexico's dominant social platform — **93.2% of Mexican internet users** are on Facebook, Statista and for many small businesses, their Facebook page IS their website. Instagram reaches **48.8 million users**. Both platforms are critical sources for WhatsApp links and business contact data.

Instagram scraping via Apify costs approximately **$2.30–$2.60 per 1,000 profiles** using the official Instagram Profile Scraper. The Instagram Search Scraper is particularly valuable for Mexico because it extracts place name, category, phone number, address, GPS coordinates, and external URLs (including wa.me WhatsApp links). Budget approximately **$230–$450 for 100K profiles** on Apify's Scale plan ($199/month). PhantomBuster is **not cost-effective at scale** — at $0.04–$0.07 per profile, it's 4–7x more expensive and limited to 300 hours of execution time even on the $439/month Team plan.

Finding Mexican business accounts works through hashtag searches (#pymesmexico, #hechoenmexico, #emprendedoresmexico, plus city-specific tags) and location-based searches. Instagram's anti-scraping measures are the most aggressive of any platform, with multi-layered blocking including TLS fingerprinting and behavioral analysis. Google Sites Expect **5–10% failure rates** even with good tools.

Facebook Pages scraping extracts page name, email, phone number, physical address, website, Messenger link, likes/followers, rating, and crucially, **ad running status** — a direct indicator of marketing budget. On Apify's Scale plan, budget approximately **$200–$500 for 100K pages**. The limiting factor is throughput: ~16,000–20,000 pages per month on the Scale plan.

Meta Ads Library scraping identifies businesses actively spending on advertising — the strongest signal that a business has budget for AI tools. A critical limitation: Meta's official Ad Library API **does not return data for Mexican commercial ads** (only EU and political ads). You must use scraping tools like Apify's "Facebook Ad Library Scraper" at **$0.75–$5.00 per 1,000 ads**. The Ad Library reveals advertiser name, ad creative, landing page URL, platforms, and start date Superads — but **not contact information or spend amounts** for Mexico. The pipeline is: scrape Ad Library → extract Page IDs → feed into Facebook Pages Scraper → get contact data.

Reviews analysis beats phone calls for finding non-answering businesses

Two approaches exist for identifying businesses that don't answer their phones: actually calling them, or analyzing reviews. The review-based approach is dramatically cheaper and legally safer.

Twilio call detection works but is expensive at scale. Outbound calls to Mexico cost **$0.016/minute (landline) or $0.0473/minute (mobile)**, Callin plus **$0.0075 per call** for Answering Machine Detection. twilio At 20 seconds average ring time across 500K numbers (60% landline, 40% mobile), the total reaches approximately **$8,500**. Twilio's API provides granular status callbacks: `no-answer`, `busy`, `failed`, `completed`, plus AMD distinguishes human answers from voicemail. At 5 calls per second, throughput is 15,000–20,000 calls per day. Retell AI is overkill — it's designed for conversational AI agents, not silent call testing.

The hybrid review analysis approach costs under $50 and delivers 85%+ accuracy. First, scrape 5 recent reviews per business (2.5M reviews for 500K businesses at **$1,250–$2,700** using Apify or Outscraper). Then apply a two-stage filter:

1. **Keyword matching (free)**: Regex against Spanish phrases — "no contestan," "no responden," "tardaron en contestar," "imposible comunicarse," "teléfono fuera de servicio." Processes 5M reviews in minutes.
2. **Claude Haiku Batch on flagged reviews (~$25–50)**: Send 1–2 star reviews to Claude Haiku via Batch API at $0.0001 per review with 50% batch discount. DEV Community Catches nuanced complaints the keyword filter misses.

Additional non-calling signals: businesses with no Google Maps "Popular Times" data, no web presence, no recent reviews (>6 months stale), and Facebook pages showing "Typically responds in days." Cross-referencing multiple signals creates a high-confidence score for non-responsiveness.

Mexico's legal framework favors business data scraping with careful execution

The LFPDPPP (updated March 2025) Cookie-Script Hogan Lovells protects personal data of natural persons but **does not apply to data of legal entities (personas morales)**. Scraping business names, commercial addresses, and published business phone numbers of incorporated companies falls outside the law's scope. However, sole proprietor data (personal names, personal phones) is protected and requires consent — tacit consent suffices for non-sensitive data White & Case LLP if a privacy notice is provided. talk-q CookieYes

Penalties are serious: administrative fines (Bitraser) up to **~$1.8 million USD** and criminal sanctions of 3 months to 5 years imprisonment for deceitful processing. (Bitraser) The enforcement body shifted from the dissolved INAI to the new Secretaría Anticorrupción y Buen Gobierno. (Pag) (Compliance Hub Wiki)

REPEP (Mexico's Do Not Call registry) specifically targets marketing/advertising calls. Silent test calls to verify if businesses answer their phones are a gray area — technically not "publicidad," but systematic automated calling could attract regulatory scrutiny. REPEP focuses on consumer protection, and most registered numbers are personal/residential. Business commercial lines are less likely to be registered. Fines for REPEP violations reach up to **~$112,000 USD**, with potential 90-day suspension of operations.

WhatsApp cold messaging to scraped numbers carries the highest risk. Meta actively monitors for spam patterns, and unsolicited bulk messages trigger permanent account bans with no recovery path. The only compliant approach is using the official WhatsApp Business API with **opted-in contacts only**. Marketing messages to Mexico cost approximately **$0.025–$0.036 per message** through the API. New accounts start with a limit of 1,000 unique recipients per day, scaling to unlimited as trust builds.

Activity	Legal risk	Key concern
Scraping Google Maps business data	**Low**	Not personal data for corporations
Scraping individual proprietor data	**Medium-High**	LFPDPPP requires consent
Automated calls to test answering (no pitch)	**Medium**	Gray area under REPEP
Marketing calls without REPEP check	**High**	Clear violation, fines to $112K
Cold WhatsApp to scraped numbers	**Very High**	Permanent Meta ban + LFPDPPP risk
Storing commercial business data	**Low**	Outside LFPDPPP for legal entities

Best compliance practices: Maintain a published Aviso de Privacidad, purchase and scrub against REPEP lists before any outbound calls, use WhatsApp Business API only with opt-in contacts, separate business entity data from individual personal data in your database, and document all data sources and processing purposes.

The recommended tech stack costs $75/month to run 5 million rows

Supabase Pro + Medium compute ($75/month) comfortably handles 5M rows with no row limits. A 5M-record database with full indexing requires approximately 5GB of storage — within

the Pro plan's 8GB quota. PostgreSQL's `pg_trgm` and `fuzzystrmatch` extensions (both available on Supabase) enable fuzzy name matching for deduplication directly in the database.

The optimal schema uses three tables: `business_leads` (canonical records with 20+ fields), `lead_sources` (tracking each data source per lead with full raw JSONB data), and `enrichment_log` (history of enrichment attempts). Deduplication runs in three passes: exact match on Google Place ID, exact match on phone number, then fuzzy match on business name + city using `similarity()` with a 0.7 threshold. Neon

n8n self-hosted (Community Edition) is the best orchestration choice at **$10–20/month** for VPS hosting, with unlimited workflow executions and native nodes for Supabase, Apify, and Outscraper. Make.com is the no-code alternative at **$130–338/month** during the initial load phase. make Zapier is 13x more expensive than Make.com at this volume and should be avoided. First AI Movers Get AI Perks

The pipeline architecture flows: n8n scheduler triggers scraping jobs → Outscraper/Apify run and return data via webhooks → Python scripts clean, normalize, and deduplicate → bulk upsert into Supabase via direct PostgreSQL connection → enrichment runs in subsequent passes. Python scripts handle the heavy lifting (bulk inserts of 500–1,000 rows per batch using `INSERT ... ON CONFLICT`) while n8n orchestrates scheduling and monitoring.

Complete cost breakdown for 3M+ Mexican business leads

One-time scraping costs

Component	Tool	Estimated cost
Google Maps businesses (3M)	Outscraper	$3,200
Google Maps contacts enrichment (3M)	Outscraper	$3,000
Google Maps reviews (2.5M)	Outscraper	$2,700
DENUE/INEGI database (5M+)	INEGI API	**$0 (free)**
Instagram profiles (100K)	Apify	$230–$450
Facebook Pages (100K)	Apify	$200–$500
Meta Ad Library (50K ads)	Apify	$38–$250
Doctoralia (180K)	Apify	$100–$200
AI review analysis (hybrid)	Claude Batch API	$25–$50

Component	Tool	Estimated cost
Subtotal (scraping)		**$9,500–$10,350**

Optional: call detection

Component	Tool	Estimated cost
Test 500K phone numbers	Twilio	~$8,500
Phone number rental (10 numbers)	Twilio	$63/month

Monthly recurring infrastructure

Component	Tool	Monthly cost
Database	Supabase Pro + Medium compute	$75
Orchestration	n8n self-hosted (VPS)	$15
Apify subscription	Scale plan	$199
Data enrichment	Apollo Basic or Hunter Starter	$49
Monthly total		**$338**

Weekly refresh scraping (ongoing)

Component	Frequency	Monthly cost
Re-scrape 50K records/week (Outscraper)	Weekly	$200
Instagram/Facebook updates	Weekly	$50–$100
Refresh total		**$250–$300/month**

Grand totals

- **One-time setup + initial scrape**: $9,500–$10,350 (without call detection) or $18,000–$19,000 (with call detection)
- **Monthly ongoing**: $588–$638

- **Total first month all-in**: $10,100–$19,650

Two-week implementation timeline

Week 1: Foundation and primary data collection

Days 1–2: Infrastructure setup

- Create Supabase project, deploy schema with all three tables, enable `pg_trgm` and `fuzzystrmatch` extensions, create indexes
- Set up n8n on a VPS (DigitalOcean/Hetzner $10–20/month droplet)
- Register for Outscraper, Apify, and INEGI DENUE API accounts
- Write Python data processing scripts for cleaning and normalization

Days 3–5: Launch primary scrapes

- Start Outscraper Google Maps scrape: 25 categories × top 100 cities (Tier 1 and 2). This will run for 2–4 days in the background and capture an estimated **1.5–2M businesses**
- Begin DENUE API data pull (Python script, free). Process all 5M+ records, extract those with phone numbers and emails. Estimated **500K–800K records with contact data**
- Configure n8n workflows for Outscraper webhook → Python processing → Supabase upsert

Days 5–7: Social media and directories

- Launch Apify Instagram Search Scraper for top 50 Mexican cities with business keywords
- Launch Apify Facebook Pages Scraper for high-value categories (restaurants, clinics, salons, gyms)
- Start Doctoralia scraping via Apify actor (all specialties, all cities)
- First batch of Google Maps data arrives — process and load into Supabase
- Run initial deduplication pass
- **Expected database size by end of Week 1: 500K–1M deduplicated leads**

Week 2: Scale, enrich, and operationalize

Days 8–10: Expand coverage

- Launch Outscraper scrape for remaining 400+ smaller municipalities
- Launch Meta Ad Library scraping to flag active advertisers

- Process and load all Instagram, Facebook, and directory data
- Run cross-source deduplication (merge Google Maps + DENUE + social media records)
- Begin review scraping for top 500K businesses with reviews

Days 11–12: Enrichment and analysis

- Run hybrid review analysis (keyword match + Claude Haiku) to flag non-answering businesses
- Start email enrichment via Hunter.io or Apollo for businesses with domains
- Cross-reference ad library data to tag businesses with active ad spend
- Build Supabase views for lead scoring and segmentation

Days 13–14: Pipeline automation and testing

- Finalize n8n automation workflows for weekly refresh cycles
- Set up monitoring and alerting (Slack notifications for pipeline failures)
- Test full pipeline end-to-end with a small batch
- Document the system and create runbook
- **Expected database size by end of Week 2: 2–3M deduplicated leads, with 500K+ having phone numbers**

What's realistic and what's not

Achievable in 2 weeks: 2–3M business records in Supabase with basic contact data (name, address, phone, category, rating). DENUE data loads in hours. Google Maps scraping of major cities completes within days. Social media scraping runs concurrently.

Stretch goals that may slip to Week 3: Complete coverage of all 500+ municipalities, full review analysis at scale, comprehensive Instagram/Facebook enrichment, and Meta Ad Library tagging. These are parallelizable but depend on scraping tool throughput and anti-bot delays.

Not achievable in 2 weeks: Testing 500K phone numbers via Twilio (throughput of ~20K calls/day means 25 business days), complete email enrichment of 3M records through Clay or Hunter.io, or building a polished front-end for the lead database. Plan these as rolling processes that continue after the initial 2-week sprint.

The critical insight for Mexico specifically: **the phone number IS the WhatsApp number** for the vast majority of small businesses. Mexican SMEs overwhelmingly use WhatsApp Business on their main business line. Rather than investing in separate WhatsApp discovery, validate phone numbers from Google Maps and DENUE, then use the WhatsApp Business API's contact check

to confirm WhatsApp availability — a far cheaper and more reliable approach than scraping Instagram bios for wa.me links.

Conclusion

This pipeline is not just feasible — it's remarkably cost-efficient for the scale involved. The combination of DENUE's free 5M-record API with Outscraper's $1-per-thousand pricing IGLeads creates a foundation that no other market offers this cheaply. The total infrastructure runs on less than $650/month after the initial scraping investment. The key strategic decisions are: use Outscraper over Apify for Google Maps (4–9x cheaper), always start with DENUE (free and comprehensive), skip traditional B2B enrichment tools like Clay and Apollo (they're built for US/EU SaaS companies, not Mexican taquerías), and never cold-message WhatsApp numbers (the ban risk is existential to the outreach operation). The businesses running Meta ads represent the highest-value segment — they've already proven willingness to spend on marketing, PowerAdSpy making them the ideal first targets for AI voice and chat agent sales.